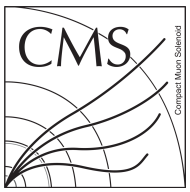
Internal review draft – 22 October 2026
No CMS DP number assigned




The Compact Muon Solenoid Experiment
CMS Performance Note




# MCP-referenced timing calibration of CMS BTL prototype bars

CMS

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**Abstract**

A timing-calibration procedure is studied using CMS barrel timing layer (BTL) prototype scintillator bars in the September 2025 beam test. A microchannel-plate photomultiplier (MCP-PMT), digitized by an oscilloscope, provides an external arrival-time reference. A trigger link identifies coincident events in the BTL readout and the oscilloscope. Segment-dependent clock alignment and a locally weighted smoothing correction are used to map the reference into the BTL time coordinate.

The dependence of the measured arrival time on pulse charge is described by quadratic functions of inverse charge for four silicon-photomultiplier channels on two reference bars. These functions provide an empirical time-walk calibration over the measured charge range. Clock-corrected profiles provide the reference for the companion dual-threshold study; archived fits are retained to document the change in calibration procedure. Timing-difference distributions illustrate the validation strategy and the need to distinguish intrinsic reference-detector resolution from residual synchronization effects. This draft documents the available results and identifies the remaining selection, uncertainty and reproducibility checks before publication.

Draft v1 for scientific discussion. Not approved for public release.
The following slides retain explicitly marked review items.

# MCP-referenced timing calibration of CMS BTL prototype bars

CMS

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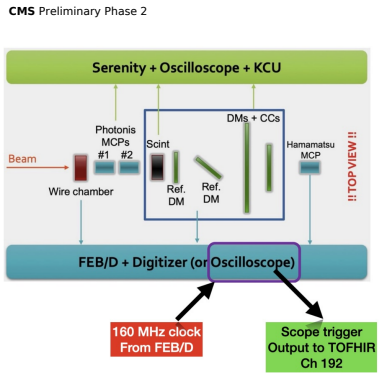
Discussion draft; not approved for public release.

## Abstract

A timing-calibration procedure is studied using CMS barrel timing layer (BTL) prototype scintillator bars in the September 2025 beam test. A microchannel-plate photomultiplier (MCP-PMT), digitized by an oscilloscope, provides an external arrival-time reference. A trigger link identifies coincident events in the BTL readout and the oscilloscope. Segment-dependent clock alignment and a locally weighted smoothing correction are used to map the reference into the BTL time coordinate.

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# Beam-test setup and the reference signal



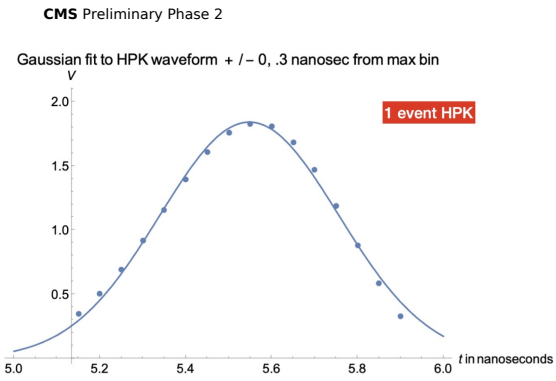

The September 2025 beam test used scintillator bars read out at both ends by silicon photomultipliers (SiPMs). A downstream MCP-PMT provides a fast reference for a restricted region of the bars.

The BTL front-end readout and oscilloscope use different time coordinates. The scope trigger reaches TOFHIR channel 192; a separate 160 MHz clock link can also be recorded by the scope.

The highlighted lower readout path is used for the reference-bar calibration.

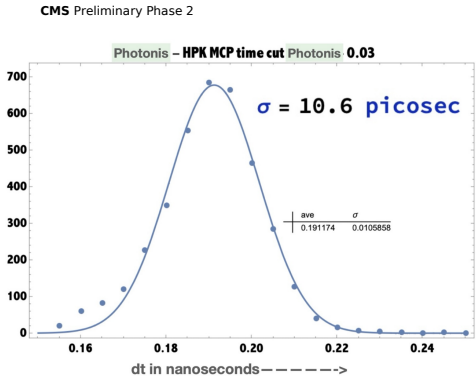
Legacy setup illustration. Detector/readout context: Refs. [1, 2]; run-specific setup from this study.

# Reconstructing the MCP pulse time



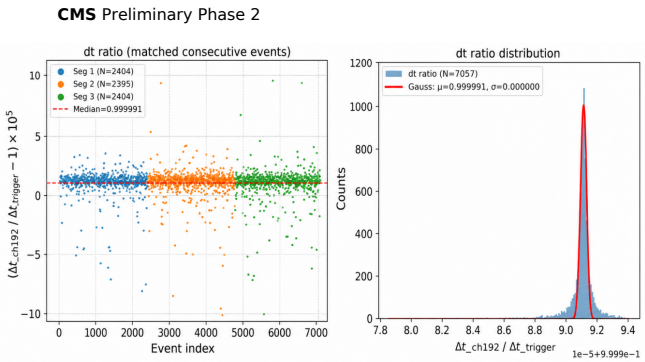

An example Hamamatsu MCP waveform is shown with its fitted pulse template. The fit estimates the pulse centroid rather than using the threshold-crossing time alone, reducing sensitivity to pulse amplitude.
The scope-trigger time identifies the event; the fitted peak supplies its relative reference time. Their difference is measured within the same oscilloscope waveform.

# Intrinsic reference timing is a separate measurement




The distribution compares the fitted times of two MCP-PMTs recorded by the oscilloscope. The measured width is 10.6 ps.
Assuming independent errors and a 7.5 ps PHOTONIS contribution gives $\sigma_{\mathrm{HPK}} \simeq \sqrt{10.6^2 - 7.5^2} = 7.5$ ps.
This scope-only estimate does not include synchronization to the BTL readout.

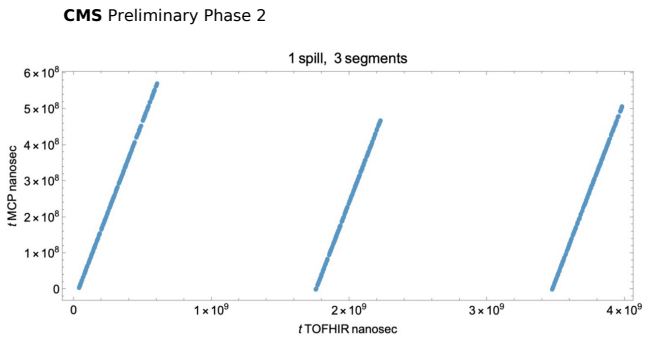
# Matching the same particles between readouts




Consecutive trigger intervals measured by channel 192 are compared with those recorded by the scope. Correlated intervals identify candidate matches despite the independent clocks.
The interval-ratio distribution tests matching consistency. It is not a measurement of the picosecond resolution of the reconstructed particle time.
Only uniquely matched events enter the reference calibration.

Implementation (repository-relative):
`FEBD_Sync/simple_sync.py`
`dedup_triggers`: duplicate removal.
`match_cumulative_time`: matching and tolerance.

# Clock-rate alignment within acquisition segments

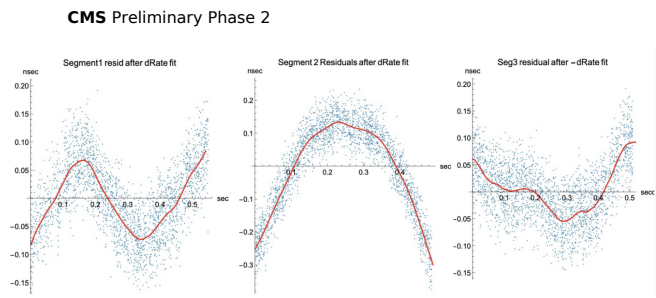



Matched scope-trigger times are compared with TOFHIR trigger times. A nearly linear relation describes the overall clock-rate difference within an acquisition segment.

An offset and a slope are needed: a slope close to one is not sufficient over a long time interval. Clock resets require separate segment coordinates.

Legacy clock-relation plot. This plot illustrates synchronization, not charge-dependent time walk.

# Correcting smooth residual clock drift




After the linear clock-rate correction, the time residual still changes within each of three segments. Points are matched events; red curves show the smooth trend estimated by LOWESS, a locally weighted regression [3].
The fitted trend is added to the linear prediction to reconstruct the expected channel-192 trigger time. This expected time is subtracted once from the SiPM time in the reference difference.

Legacy residuals in ns versus segment time in seconds. These are pre-LOWESS residuals, not final timing resolutions. Implementation: `TimeCalibration/ch192_vs_trigger_lowess.py`, lines 657–668 and 816.

## A single, explicit time convention

Let $g(u)$ map a scope-trigger timestamp $u$ into the TOFHIR time coordinate, including the linear clock relation and the fitted smooth drift. Define the within-waveform difference

$$\delta_{\mathrm{MCP}} = t^{\mathrm{scope}}_{\mathrm{peak}} - t^{\mathrm{scope}}_{\mathrm{trigger}}.$$

The reference difference implemented in the calibration is

$$\Delta t_i = t_i^{\mathrm{TOFHIR}} - g(t^{\mathrm{scope}}_{\mathrm{trigger}}) - \delta_{\mathrm{MCP}},$$

up to a fixed cable/geometry offset and the negligible-rate-conversion approximation for the short within-waveform interval.

The charge-dependent component of $\Delta t_i$ is the time walk. An arbitrary constant does not change a shape calibration.

**Implementation:** `TimeCalibration/ch192_vs_trigger_lowess.py`, lines 816 and 1102. Both per-file and combined-run calculations subtract (`mcp_peak - mcp_trig`).

## Reference bars and calibration selection

| Upstream bar | SiPM channels | Role |
|---|---|---|
| 7 | 133, 154 | First reference pair |
| 8 | 137, 150 | Second reference pair |

- Select coincident SiPM hits and a unique MCP-associated event, using channel 192 as the TOFHIR indicator.
- Require a successfully reconstructed MCP pulse and consistent peak-to-trigger timing. This is a waveform-quality cut, not a particle transit-time measurement inside the MCP.
- Use $Q$, the TOFHIR ADC charge proxy. It is not a deposited energy in MeV.
- Fit the profile of $\Delta t_i$ versus $1/Q$ over the retained charge range.

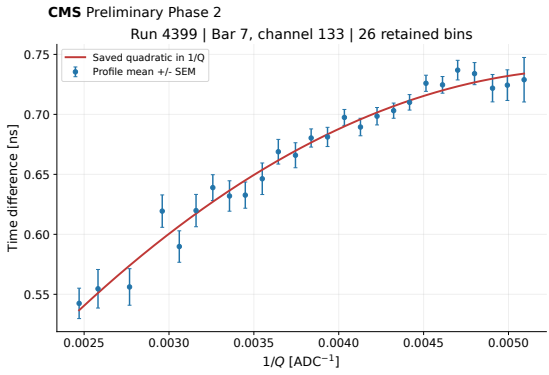
**Implementation (repository-relative):**

`TimeCalibration/ch192_vs_trigger_lowess.py`: coincidence and walk-sample selection.

`TimeCalibration/ch192_vs_trigger.py`: `build_mcp_map`, optional MCP cuts.

`TimeCalibration/bar_processing.py`: peak-to-trigger timing selector.

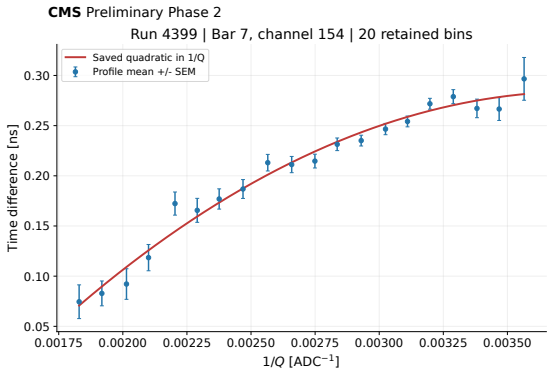
# Archived reference: bar 7, channel 133




Run 4399 events are grouped in inverse-charge bins. Each point is the mean time difference; the vertical bar is its saved standard error of the mean (SEM). The curve is the saved binned quadratic $W(Q) = a + b/Q + c/Q^2$. There are 26 retained profile points. The time coordinate includes the historical constant display offset. These archived means use measured channel-192 timing. An unweighted fit to the means reproduces the saved coefficients; SEM bars are not fit weights.

Redrawn from archived profile and coefficient CSVs. No refit. Saved display convention includes a +26 ns shift.

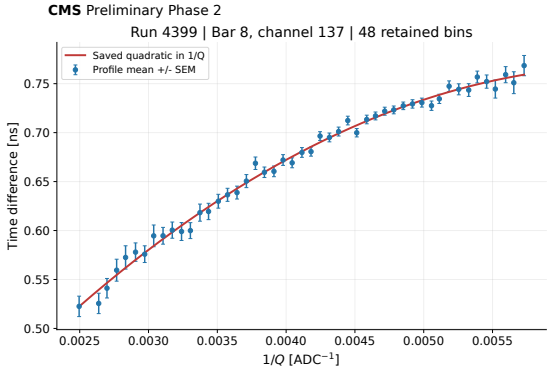
# Archived reference: bar 7, channel 154




The opposite end of bar 7 is calibrated independently. Points show the mean time difference and SEM in 20 retained inverse-charge bins.
The red curve is the saved quadratic in $1/Q$, not a quadratic in $Q$.
Channel-dependent coefficients allow for different pulse responses at the two bar ends. The archived fit is unweighted and uses measured channel-192 timing. The retained bin counts are channel dependent.

Run 4399 legacy calibration, redrawn without refitting. Same +26 ns display convention as channel 133.

# Archived reference: bar 8, channel 137

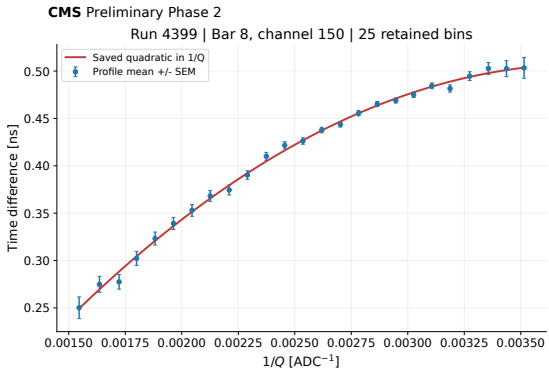



The channel-137 time difference increases as inverse charge increases: smaller signals cross a fixed threshold later.

Points are profile means with SEM; the red line is the saved binned quadratic. There are 48 retained points. The correction is applicable over the measured charge support, not automatically outside it. This unweighted, measured-channel-192 result is retained for comparison with the clock-corrected reference shown next.

Run 4399 legacy calibration. Coefficients and numerical points are included with the draft.

# Archived reference: bar 8, channel 150




The opposite end of bar 8 has its own charge-dependent delay. The 25 points show the mean timing difference and SEM; the curve is the saved binned quadratic.

The four curves demonstrate the calibration procedure. Different numbers of retained points are preserved rather than imposing an undocumented common binning. This is the archived, unweighted fit using measured channel-192 timing.

Run 4399 legacy calibration. Coefficient units are ns, ns ADC and ns ADC$^2$ for $a$, $b$ and $c$.

## Clock-corrected reference shared with Note 2

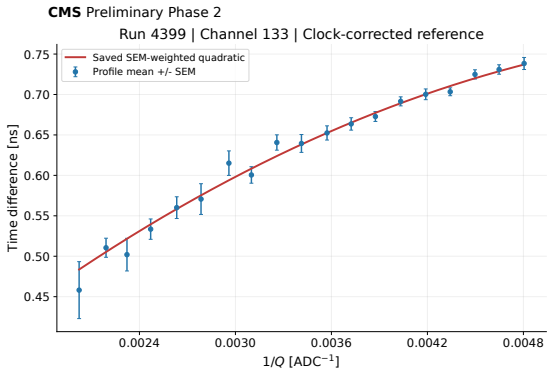
The following results use the expected trigger time $g(u)$ reconstructed from the segment-wise linear clock relation and LOWESS drift:

$$\Delta t_i = t_i^{\mathrm{TOFHIR}} - g(t_{\mathrm{trigger}}^{\mathrm{scope}}) - (t_{\mathrm{peak}}^{\mathrm{scope}} - t_{\mathrm{trigger}}^{\mathrm{scope}}).$$

- Run 4399: exactly five TOFHIR hits, comprising the target upstream pair, downstream channels 234 and 243, and channel 192, with a unique MCP match.
- Require finite, positive charge and slope-related quality quantities; retain $Q$ at or above the left half-maximum boundary of the spectrum peak.
- Form 20 uniform $1/Q$ bins; retain times within $5(1.4826\,\mathrm{MAD})$ of each bin's median, with at least five entries. Bars show the SEM after clipping.
- Fit $W(Q) = a + b/Q + c/Q^2$ with an objective weighted by $1/\mathrm{SEM}^2$. The next four figures use the exact saved reference coefficients used in Note 2.

Source snapshots: `2026-10-22_clock_quality/inputs`. The displayed $+26$ ns constant changes neither the shape nor the saved coefficients.

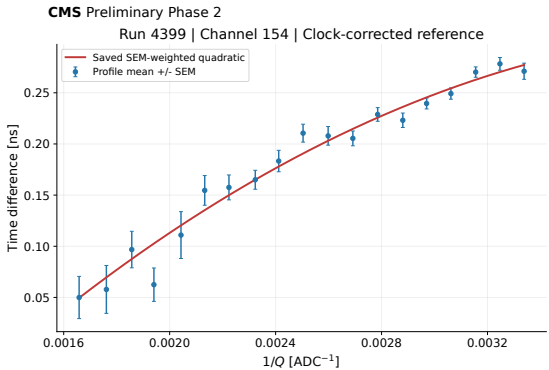
# Clock-corrected reference: bar 7, channel 133




The time difference now uses the reconstructed trigger time rather than the measured channel-192 timestamp, suppressing the fitted clock drift.
Points show mean time differences with SEM in 19 retained inverse-charge bins.
The red curve is the saved SEM-weighted quadratic used in Note 2.
The curve is shown only within retained-bin support and the outer profile means. Missing bins are not extrapolated across.

Run 4399; frozen Note2 input, redrawn without refitting. A constant +26 ns shift is applied only for display.

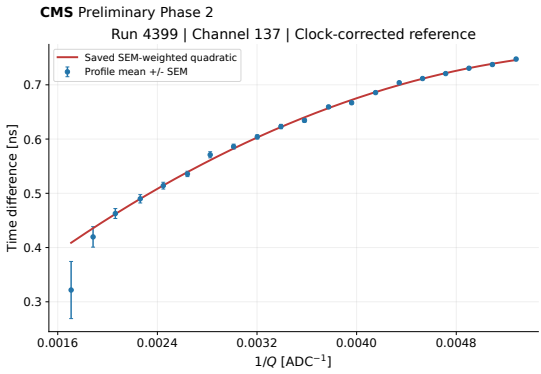
# Clock-corrected reference: bar 7, channel 154




The opposite end of bar 7 is calibrated with the same clock-reconstruction and profile procedure. Its independent coefficients describe its own charge-dependent timing response. The 19 points are retained-bin means; vertical bars show SEM after timing clipping. The red curve is the saved uncertainty-weighted quadratic in $1/Q$.

Exact reference snapshot used in Note2. Coefficients and profile points are included with this note; the +26 ns display shift is separate.

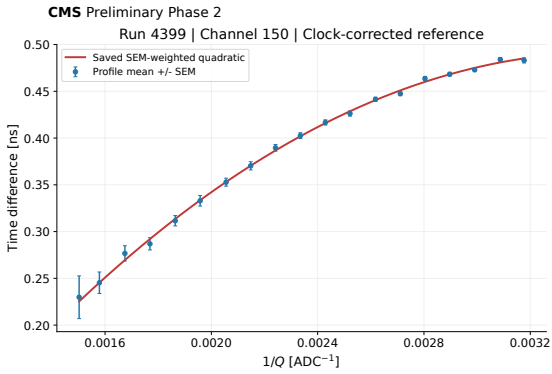
# Clock-corrected reference: bar 8, channel 137




There are 20 retained inverse-charge bins for this channel. Smaller charge gives a larger timing delay, as expected for fixed-threshold readout of pulses with different amplitudes.
Points and bars are the mean and SEM after clipping. The saved quadratic is the MCP benchmark for the channel-137 inverse-slope comparison in Note 2.

Run 4399; expected-trigger timing; refitted after clock-quality selection. The +26 ns shift affects only the vertical display origin.

# Clock-corrected reference: bar 8, channel 150




The channel-150 reference uses 19 retained profile bins and the same clock-corrected time definition as the other three channels.
Bars represent SEM, not the spread of individual event times. The red line is the saved SEM-weighted inverse-charge quadratic. These four results connect the MCP calibration in this note to the AWC shape comparison in Note 2.

Run 4399; coefficients are preserved in their original ns convention. Only the figure receives the constant +26 ns display shift.

## Archived and clock-corrected shapes are close

Compare both saved functions on jointly populated charge support, after removing a constant offset only. No multiplicative scaling is fitted.

| Channel | Shape RMS difference [ps] | Maximum absolute difference [ps] |
|---|---|---|
| 133 | 3.20 | 10.63 |
| 154 | 4.29 | 9.68 |
| 137 | 1.33 | 3.22 |
| 150 | 4.91 | 10.47 |

The current reference includes finite-timestamp reset handling and a clock-only quality requirement. The archived and current versions also differ in trigger reconstruction, selection, clipping and fit weights.

These are differences between calibration functions, not timing resolutions or a statistical-equivalence test. Note 2 uses the clock-corrected version.

# Clock quality before timing validation

Consecutive finite scope timestamps define segment boundaries. Missing timestamps cannot conceal a reset; nonfinite triggers have no valid mapping.

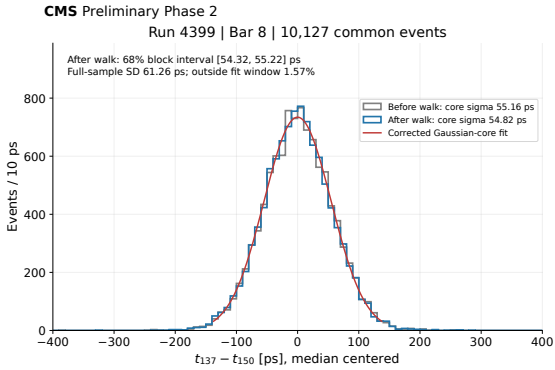
For each topology, file and segment, fit channel 192 against the scope trigger. Reject the whole group if the line-residual scale $1.4826\,\mathrm{MAD}$ exceeds 1 ns, before LOWESS. This rule uses clocks only, not SiPM arrival times.

Six finite-trigger bar-8 groups fail. Changing the threshold from 1 ns to 1 $\mu$s retains the same groups. Suspected one-event association offsets are excluded, not repaired by shifting MCP rows.

Refit the channel walk curves after this selection. For the following timing comparison require both ends to pass quality, peak-left charge cuts and their own populated reference support within the outer profile means. No additional selection is made on either timing-difference observable.

Implementation: `Q_Walk/qwalk_t0_data.py`; new references: `2026-10-22_clock_quality`.

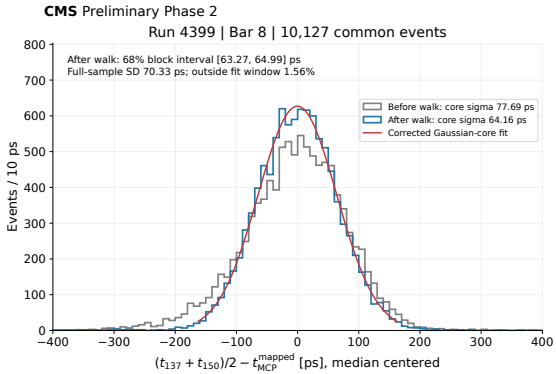
# Timing validation using the two bar ends




The same 10127 bar-8 events are shown before and after subtracting the two channel-specific walk functions. The common clock mapping is fixed. The Gaussian-core width changes from 55.16 to 54.82 ps. The corrected full-sample SD is 61.26 ps; 1.57% lie outside the core window.
The 16th–84th percentile interval is [54.32, 55.22] ps from 400 file/segment-block bootstrap replicas, with calibrations held fixed. Ref. [4] gives the general resampling principle.

Truncated-normal likelihood within median $\pm 2.5(1.4826\,\mathrm{MAD})$. Each histogram is median-centered for display only; the SD includes all events.

# Timing validation against the MCP reference




The bar-average time minus mapped MCP time uses the identical event sample and clock reconstruction before and after channel-wise walk correction. The core width changes from 77.69 to 64.16 ps. The corrected full-sample SD is 70.33 ps; 1.56% lie outside the fit window.

The conditional block-bootstrap interval is [63.27, 64.99] ps. This observable includes detector, MCP and residual clock contributions.

Same core-fit definition as the end difference. The display range does not restrict the SD. No segment-by-segment timing recentering is applied.

## What these timing widths establish

The reproduced distributions establish conditional before/after performance, not intrinsic bar resolution or a separate synchronization-jitter measurement. At fixed position, for end timing errors $e_L, e_R$,

$$\mathrm{Var}\Big(\frac{e_L + e_R}{2}\Big) = \frac{\mathrm{Var}(e_L - e_R)}{4} + \mathrm{Cov}(e_L, e_R).$$

Halving an end-difference width requires controlled position effects and negligible end-error covariance. Shared MCP and clock errors also prevent inferring detector covariance directly from the measured reference residuals.

The two separately fitted core widths are not variances of a common model. Their quadrature difference is not a justified synchronization-jitter estimate. This version reports same-event timing validation with conditional statistical intervals. Intrinsic bar resolution and synchronization-jitter extraction are outside its scope. Position, covariance, segment-offset and calibration systematics are not included in the quoted intervals.

## Reference-calibration result and scope

- A trigger link permits common-event calibration across independent BTL and oscilloscope time coordinates.
- A linear clock relation plus a smooth residual correction defines the reference-time mapping.
- Four clock-corrected inverse-charge profiles and quadratic functions provide the reference used in Note 2. The archived fits remain as a comparison; numerical points, SEM and both coefficient sets accompany this draft.
- Intrinsic MCP timing, reference-bar timing and cross-system synchronization are different performance quantities.
- Charge range and operating conditions limit transfer to other samples; a constant time offset is not a walk-shape difference.

For the same selected events, walk correction reduces the bar–MCP core width from 77.69 to 64.16 ps. This is a conditional timing-validation result, not an intrinsic-resolution measurement or a complete uncertainty budget.

# References

[1] CMS Collaboration, *A MIP Timing Detector for the CMS Phase-2 Upgrade*, CERN-LHCC-2019-003, CMS-TDR-020 (2019). CERN document record.

[2] E. Albuquerque et al., *TOFHIR2: The readout ASIC of the CMS Barrel MIP Timing Detector* (2024). arXiv:2404.01208.

[3] W. S. Cleveland, *Robust Locally Weighted Regression and Smoothing Scatterplots*, J. Am. Stat. Assoc. **74** (1979) 829–836. doi:10.1080/01621459.1979.10481038.

[4] B. Efron, *Bootstrap Methods: Another Look at the Jackknife*, Ann. Statist. **7** (1979) 1–26. doi:10.1214/aos/1176344552.

These sources provide detector/readout context and methodological background. The run-specific selections, implementations and numerical results are those of this study, not measurements or prescriptions taken from these references.

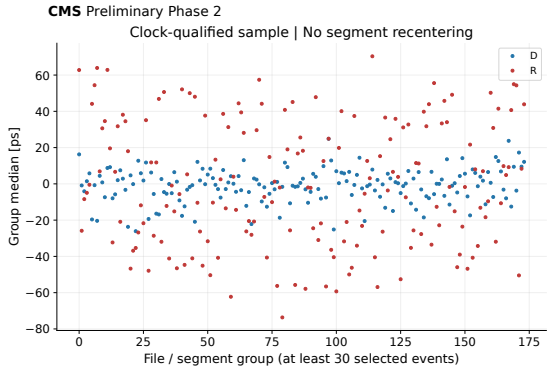
# Review queue before publication

| ID | Decision or missing evidence |
|---|---|
| N1-6 | Local citations and candidate packaging checked; conveners must confirm template, DP metadata and cover handling. |

Sources: supplied `Note1.pdf`; candidate editable manuscript `CMS-Note/cmsnote`; archived calibration CSVs in `Q_Walk/reference_quadratic_t0`.

The figure manifest records the exact copied inputs and checksums. This review slide is removed from a public release.

# Backup: residual segment offsets



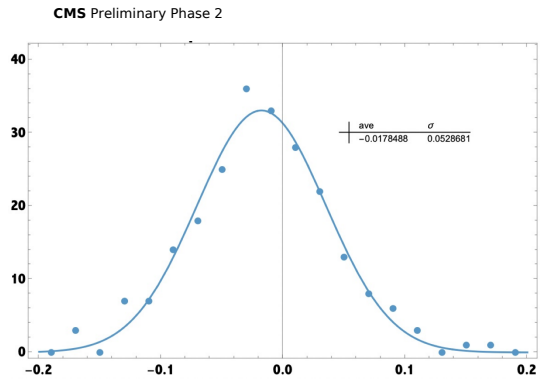

Points show median end differences ($D$) and bar-minus-MCP differences ($R$) for groups containing at least 30 selected events.

Clock-quality selection removes gross reconstruction failures, but smaller group offsets remain. They contribute to the aggregate timing distribution.

The main timing figures do not subtract these medians. Their effect must be understood before extracting a synchronization contribution.

Group medians are descriptive point estimates, not independent resolution measurements. Identifiers and event counts accompany the numerical data.

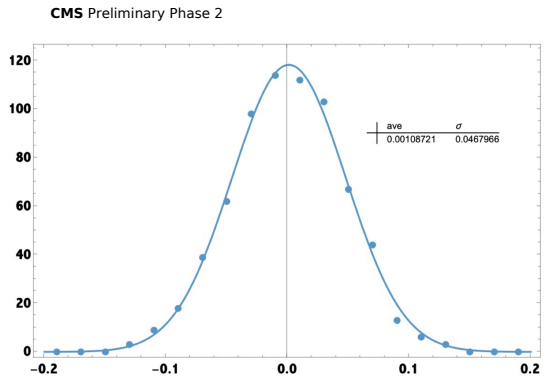
# Backup: historical end-difference illustration




This archived figure displays a width of 52.87 ps. Its exact event sample, histogram inputs and fit definition have not been recovered.
It is retained only to identify the historical result. The reproducible clock-qualified before/after comparison replaces it in the main result. No intrinsic 26 ps bar-resolution claim is made from this image.

Historical illustration, not the current calibration result.

# Backup: historical bar–MCP illustration

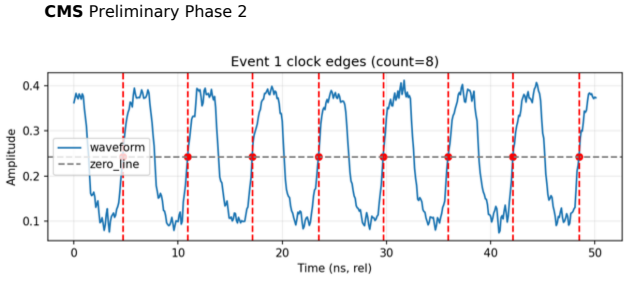



The archived panel reports 46.8 ps. Whether this denotes a fitted core width or a sample RMS, and the original selection, have not been established.

It is not combined with the current end-difference result. The new bar–MCP plot specifies its common event sample, core fit, full-sample SD and conditional statistical interval.

Historical illustration; not independently reproduced by the documented full-run pipeline.

# Backup: clock-phase reconstruction



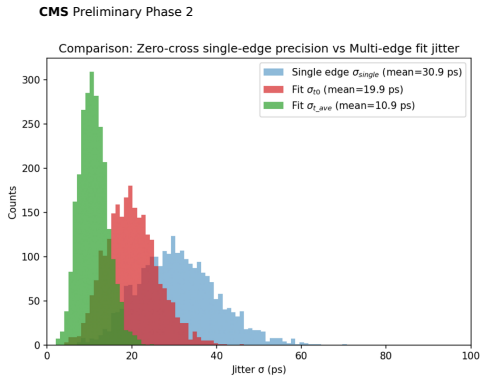

The reference clock is sampled by the oscilloscope. Several rising-edge crossings can be reconstructed within a waveform.

Fitting edge time against edge index allows the clock phase to be estimated using multiple cycles. Evaluating the fit near the mean edge index avoids unnecessary extrapolation.

This is an alternative synchronization diagnostic, distinct from the trigger-link branch used for the legacy reference curves.

Legacy illustration. Correlated edge errors prevent assuming an automatic $1/\sqrt{N}$ precision improvement.

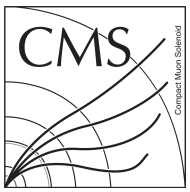
# Backup: comparing clock estimators




The recorded MCP time is compared with three clock-phase estimators: a single edge, a regression intercept and the fitted time at the mean edge index. The distributions illustrate estimator dependence. Their widths contain the contributions present in this specific measurement and are not automatically an absolute clock-accuracy measurement. The source arrays, common selections and width definitions are not established. This internal discussion figure is excluded from the publication candidate.

Legacy clock-estimator diagnostic, retained as a discussion backup.

Internal review draft – 22 October 2026
No CMS DP number assigned




The Compact Muon Solenoid Experiment
CMS Performance Note




# Dual-threshold time-walk calibration for CMS BTL prototypes

CMS

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**Abstract**

A charge-dependent time-walk model is studied for CMS barrel timing layer (BTL) prototype channels using two discriminator thresholds. Their current separation divided by the measured crossing-time separation provides a finite-interval pulse-slope proxy. A linear profile fit describes this proxy as a function of the measured charge, and an inverse-slope function models the timing correction with one amplitude parameter per channel.

Slope measurements from run 4076 are compared with independent microchannel-plate (MCP) reference calibrations from run 4399. The two runs contain different events. A free constant offset and the correction amplitude are fitted only where both measurements have populated charge support. The resulting agreement is a calibration-shape comparison, not an independent measurement of timing resolution.

The draft also examines channel-to-channel variations and an exploratory prediction of the correction amplitude from internal pulse information. Such a prediction could enable operation without an event-by-event external time reference, but its transferability and full uncertainty remain to be established.

Draft v1 for scientific discussion. Not approved for public release.
The following slides retain explicitly marked review items.

# Dual-threshold time-walk calibration for CMS BTL prototypes

CMS

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Discussion draft; not approved for public release.

## Abstract

A charge-dependent time-walk model is studied for CMS barrel timing layer (BTL) prototype channels using two discriminator thresholds. Their current separation divided by the measured crossing-time separation provides a finite-interval pulse-slope proxy. A linear profile fit describes this proxy as a function of the measured charge, and an inverse-slope function models the timing correction with one amplitude parameter per channel.

Slope measurements from run 4076 are compared with independent microchannel-plate (MCP) reference calibrations from run 4399. The two runs contain different events. A free constant offset and the correction amplitude are fitted only where both measurements have populated charge support. The resulting agreement is a calibration-shape comparison, not an independent measurement of timing resolution.

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## Two complementary runs, not the same events

| Sample | Available information | Role in this study |
|---|---|---|
| Run 4076 | Valid dual-threshold timing and charge | Measure the pulse-slope proxy $S(Q)$ |
| Run 4399 | MCP-associated reference timing and charge | Measure the empirical walk $W_{\mathrm{MCP}}(Q)$ |

The runs contain different particles and are never matched event by event. The comparison transfers a channel-response model between runs.

The upstream and downstream modules each contain bars read out at two ends. Detector/readout context is given in Refs. [1, 2]. The channel/bar mapping determines the coincidence pairs; the same channel number is compared across the two runs.

The comparison assumes the same nominal detector setup in both runs, with MCP reference timing in run 4399 and dual-threshold timing in run 4076. Cross-run response differences are not included in the quoted uncertainties.

## From two crossing times to a walk model

For thresholds $I_1 < I_2$ and their valid crossing times $t_1, t_2$, define

$$S = \frac{I_2 - I_1}{t_2 - t_1} \quad [\mu\mathrm{A/ns}], \qquad \overline{S}(Q) \simeq c_0 + c_1 Q.$$

$S$ is a finite-interval rising-edge slope proxy. It is not the current difference $\Delta I$ itself and is not necessarily the instantaneous derivative at the first threshold.

$$W_Q(Q) = \frac{A}{c_0 + c_1 Q}, \qquad t_{\mathrm{corr}} = t_{\mathrm{measured}} - W_Q(Q).$$

The amplitude $A$ (AWC) has units of current and is constant for a channel and operating configuration. The correction varies event by event through $Q$.

## Timing reconstruction and threshold scale

The run-4076 sample is file `23_e.root`. For coarse codes $C_1, C_2$,

$$d_c = (C_2 - C_1) \bmod 1024, \qquad t_2 = t_1 + (6250\,\mathrm{ps})d_c - f_1 + f_2.$$

The coarse codes use signed 64-bit arithmetic before subtraction; the coarse time term uses floating point, avoiding the old 16-bit multiplication overflow.

The fine-time lookup stores the mean of $t_1$ mod $6250\,\mathrm{ps}$ for each (channel, TAC, fine-code) key, using all input hits. The same lookup supplies $f_1$ and $f_2$. No selected hit in the 22 audited channels has a missing entry.

All selected hits have threshold register `step2=113621`, decoded as DAC codes 10 and 35. The implemented scale gives

$$\Delta I = (35 - 10) \times 0.3125\,\mu\mathrm{A} = 7.8125\,\mu\mathrm{A}, \qquad \frac{S}{\mu\mathrm{A/ns}} = 1000\,\frac{\Delta I/\mu\mathrm{A}}{(t_2 - t_1)/\mathrm{ps}}.$$

Implementation: `Q_Walk/qwalk_core.py`. This checks the implemented conversion, not an independent hardware-threshold calibration.

## Selection and profile construction

- Require one hit at each end of the target bar and no other hit in that module. The other module is unconstrained; channel 192 is ignored.
- Require finite $Q, S$, $\Delta I > 0$, $t_2 - t_1 > 0$ and $d_c = 0$. This selects the same-coarse-bin prompt population.
- Require $Q \geq Q_{\mathrm{left}}$, the left half-height boundary of the smoothed charge histogram. The Landau–Gaussian fit does not set this cut. There is no additional upper-charge cut or 100 ps minimum-rise cut.
- Form ten equal-width bins over the selected charge range. Plot mean $S$ and hit SD; fit with SEM weights. Bins with undefined or negligible SEM are omitted.
- Bootstrap selected hits with fixed bin edges and refit 1000 times. Quote the 16th–84th percentiles; Ref. [4] gives the general principle.

All selected hits are used; there is no train/test split. Implementation: `Q_Walk/qwalk_two_hit_pair_profile.py`, with the newer optional minimum-rise cut disabled.

## Reproducing the frozen run-4076 profiles

Replaying `23_e.root` reproduces the selected counts, ten-bin profiles and linear-fit coefficients for all 22 channels. Saved bootstrap coefficients reproduce their reported percentile intervals.

| Channel | Target-pair hits | Prompt hits | Peak-right hits |
|---|---|---|---|
| 137 | 25979 | 20789 | 19126 |

For channel 137, the histogram-derived boundary is $Q_{\mathrm{left}} = 184.9163$ ADC. Charge-bin centers are the mean measured $Q$, not the geometric bin centers.

The frozen selection retains rises below 100 ps in channels 130, 139, 151, 159, 236 and 238. These events were not removed retrospectively. Their effect belongs to the selection-sensitivity and channel-quality studies.

Channel 135 has one singleton bin: its SEM is undefined and only nine bins enter that channel's linear fit. No fit or plotted data changed in this audit.

Audit, input/code hashes and per-channel counts: `DPS_NOTES/review_2026-10-22_n2_2/`.

## Reference version shared with Note 1

The run-4399 MCP functions were refitted on 22 October 2026 after correcting segment detection across missing trigger timestamps and excluding invalid clock groups. Note 1 documents the timing validation and remaining limitations.

For each topology, file and segment, the scaled MAD of the channel-192 versus scope-trigger line residuals must not exceed 1 ns before LOWESS [3]. The selection uses clock information, not SiPM timing residuals. Suspect event associations are excluded; no MCP rows are shifted.

All AWC-and-offset fits in this note use these refreshed references. The run-4076 slope fits, bootstrap replicas and measured-support matching definition are unchanged. The charge spectra use the updated reference sample.

AWC intervals still hold the reference fixed. They do not account for reference-fit uncertainty, clock-selection uncertainty or cross-run transfer.

Numerical version: `Q_Walk/results/2026-10-22_clock_quality`.

## Sensitivity to the refreshed reference

The run-4076 input is fixed; these changes arise from the reference rerun, including its automatically determined charge selection and measured support.

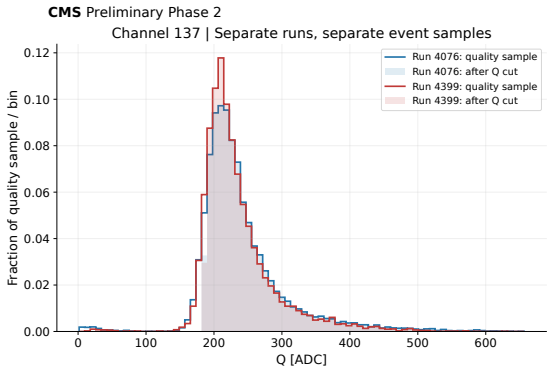
| Channel | Previous AWC [$\mu$A] | Refreshed AWC [$\mu$A] |
|---|---|---|
| 137 | 18.513 | 18.332 |
| 136 | 27.899 | -4.240 |
| 151 | 7.649 | 9.354 |

Channel 136 has only 210 selected reference hits. Its sign change, and the 22% shift for channel 151, expose sensitivity not covered by slope-only errors. They are retained in the exploratory summary, not accepted as robust calibrations.

Sparse clock groups retain the existing line-only fallback when fewer than ten alignment inliers are available. A small residual scale in such a group does not independently validate its event association.

**Review N2-6:** Require reference stability and adequate clock statistics before defining the final channel-performance sample.

# Charge spectra define the available comparison sample

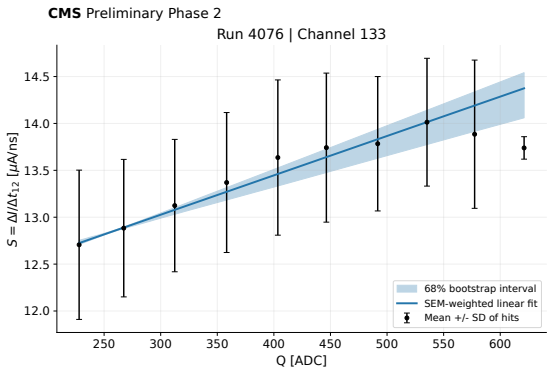



Channel 137 charge spectra are shown for the two independent runs. Lines show quality-selected hits; shaded spectra retain the additional charge-peak selection. Each run is normalized to its own quality-sample count.

The charge proxy is in ADC units, not MeV. Different populations and acceptances motivate restricting the curve alignment to jointly supported charge values.

Recovered histogram counts, not a new peak fit. Run-4076 profile selection starts at about 184.9 ADC for this channel.

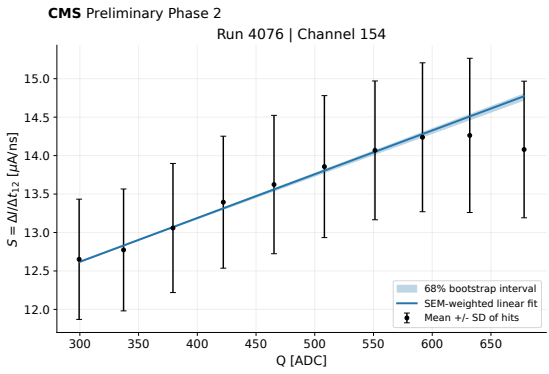
# Channel 133: fitting the slope profile




Run 4076 channel-133 hits are grouped into ten charge bins. Points show mean $S$; bars show the SD of individual selected hits. The line is the saved SEM-weighted fit $c_0 + c_1 Q$.
The shaded interval is the central 68% bootstrap interval of the fitted mean curve. It is shown at its true size, without visual enlargement.

Extended-profile snapshot. SD describes hit spread; SEM and the fit band describe different statistical quantities.

# Channel 154: the opposite end of bar 7

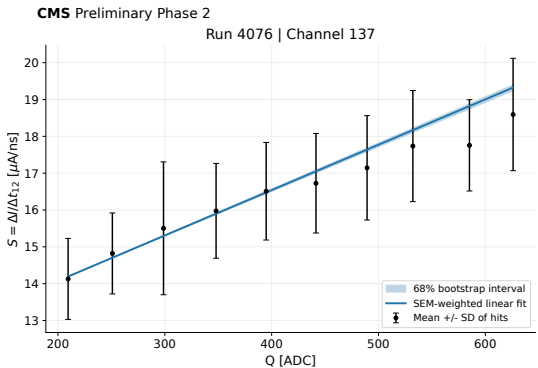



The same selection and profile procedure is applied to channel 154. The mean slope increases with charge, but the response is calibrated independently at each bar end. Bars show hit SD. The line uses SEM weights and the band propagates the saved bootstrap coefficient replicas, including their correlation.

**Review N2-3:** Linear-model adequacy remains open. The separate sensitivity audit is retained outside this note; this version keeps the original comparison.

Run 4076, extended-profile snapshot. No model refit is performed for the slide rendering.

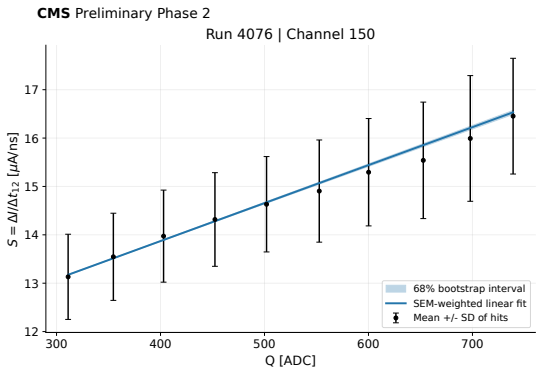
# Channel 137: pulse slope and charge




The selected run-4076 sample contains 19 126 channel-137 hits. Ten profile points constrain a linear pulse-slope model over the observed charge range. The SD bars show event-to-event spread, while SEM weights determine the fit. The narrow shaded band is the conditional 68% bootstrap interval of the mean prediction, not a single-hit uncertainty.

Selected count and fit are read from the saved extended-profile metadata. No uncertainty band is magnified.

# Channel 150: the opposite end of bar 8




Channel 150 is treated with the same profiling procedure. The charge-dependent slope provides the denominator of the proposed walk function; a separate amplitude $A$ determines the correction strength. The line and shaded interval describe the fitted mean. Points and SD bars retain information on the spread within the selected charge bins.

Run 4076, extended-profile snapshot. Constant AWC does not mean a constant timing correction.

## Determine AWC using only jointly measured support

The run-4399 reference is a fitted quadratic in inverse charge,

$$W_{\mathrm{MCP}}(Q) = a + b/Q + c/Q^2.$$

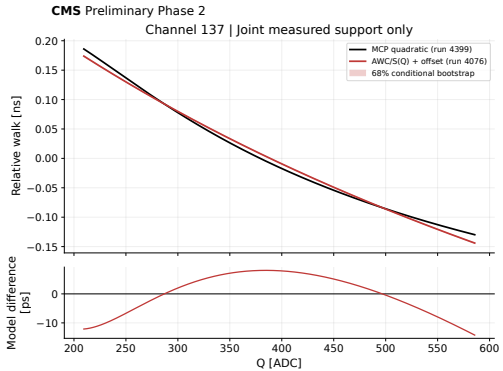
Let $\Omega$ be the intersection of the two outer profile-mean ranges, restricted to charge bins populated in both runs. Fit

$$(A, B) = \arg\min_{A,B} \sum_{Q_k \in \Omega} \left[ \frac{A}{c_0 + c_1 Q_k} + B - W_{\mathrm{MCP}}(Q_k) \right]^2 .$$

The saved implementation samples a uniform 3000-point charge grid before removing unsupported intervals. The offset $B$ absorbs the arbitrary time zero.

This is uniform-charge shape matching, not an event-weighted likelihood. It uses neither extrapolated tails nor a single central anchor point. Weighting sensitivity and correlated MCP-fit uncertainty are evaluated in the uncertainty check below. Uniform Q remains the nominal definition, not a uniquely optimal estimator.

# Channel 137: inverse-slope and MCP walk shapes

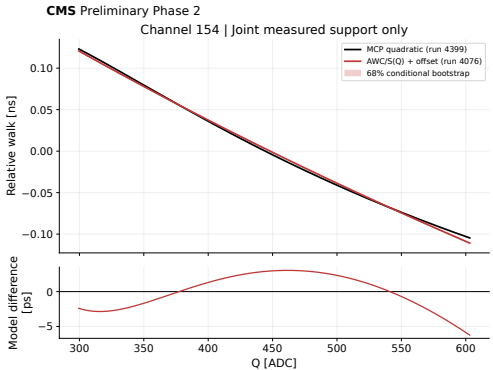



The black curve is the run-4399 MCP quadratic. The red curve is the run-4076 inverse-slope model after fitting AWC and a constant offset. Only jointly measured charge support is shown.

Lower panel: run-4076 model minus MCP model. Its RMS shape mismatch is 6.80 ps on the supported uniform-Q grid, not an event timing resolution. The original band varies the run-4076 slope fit while holding the MCP curve fixed. The later uncertainty table also includes MCP-fit uncertainty.

AWC = 18.332 $\mu$A; conditional 16–84% interval [18.120, 18.607] $\mu$A. Both curves share one display offset.

# Channel 154: a second shape-comparison example

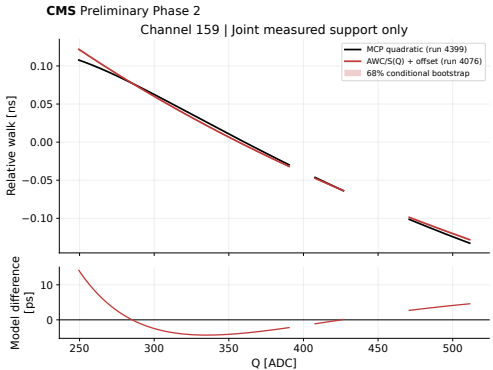



The same overlap-only objective determines AWC and the offset for channel 154. Both plotted curves are calibrated models, not matched events from different runs.

The lower panel directly displays the remaining shape mismatch, with no additional line subtracted. This measures model-shape closure, not corrected-event timing resolution.

Conditional bootstrap band: 4076 slope replicas, refitted AWC and offset, fixed 4399 quadratic.

# Channel 159: missing support is not extrapolation

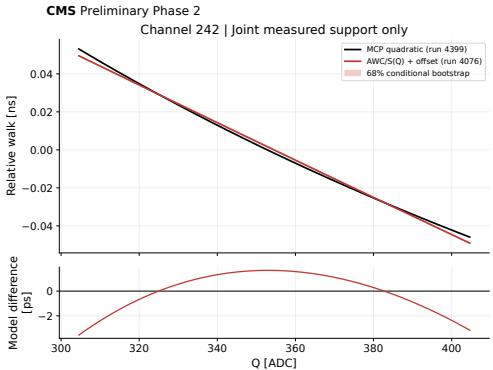



Only intervals with populated profile bins in both runs contribute to the alignment. Breaks in the displayed curves mark excluded charge intervals. They do not represent discontinuities in either analytic model.

The lower panel uses exactly the same supported points. All displayed intervals participate in the common AWC-and-offset fit. It is a model difference, not an event-time distribution.

The stored comparison excludes 703 of the 3000 initial grid points for this channel. No line is drawn through those gaps.

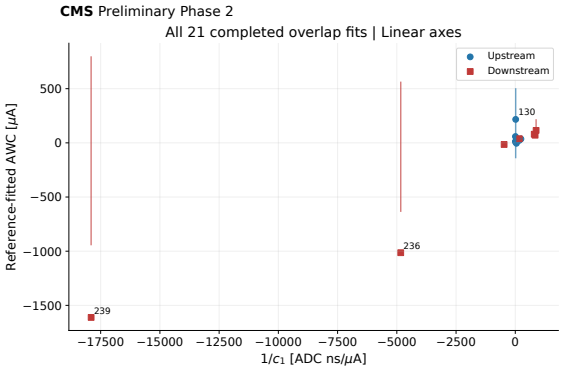
# Downstream example: channel 242




The method is also applied to a downstream channel. In run 4399, upstream bar 8 supplies the coincidence pair alongside the downstream target pair and MCP indicator.
In run 4076, the target downstream pair is required, without fixing an upstream pair. The curves use only their common measured support. The same nominal-setup assumption is used for downstream channels.
Selection-dependent response differences remain unquantified.

The lower panel is a model difference, not a timing-resolution measurement. The other module is not used as an internal timing reference to determine AWC here.

# AWC versus inverse slope coefficient: all completed fits

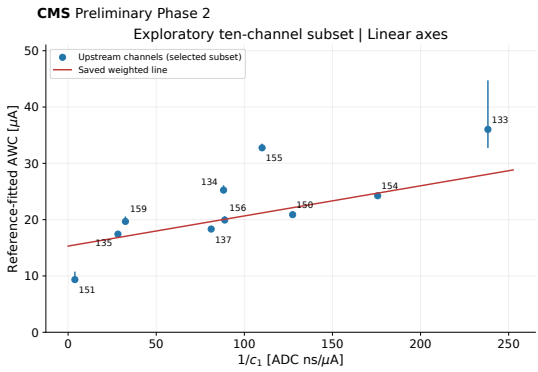



Each point is one channel, not one charge bin. Blue circles are upstream and red squares downstream. The x coordinate is $1/c_1$, the inverse coefficient of the linear $S(Q)$ fit; the y coordinate is reference-fitted AWC. Bars span the 16th–84th bootstrap percentiles. There are 21 completed fits; channel 238 has no measured overlap.

**Review N2-6:** Publish eligibility and failure criteria, including narrow support and poorly constrained channels.

Linear axes. Large intervals and negative coefficients are retained; this is not a final quality-selected performance sample.

# A selected subset does not establish a universal law




The ten-channel upstream subset previously selected for discussion is shown on linear axes. Bars are conditional AWC bootstrap intervals; channel numbers identify the points. The saved weighted straight line has $\chi^2/\mathrm{ndf} = 538.6/8$ under its limited error model. It does not adequately describe the variation.

**Review N2-6:** The subset was selected interactively. It is exploratory, not an unbiased population-level validation.

The fit treats x as fixed and uses half the 16–84% AWC span as its symmetric uncertainty. Correlations and reference errors are omitted.

## What the present uncertainties do and do not include

| Displayed quantity | Meaning |
|---|---|
| Profile bars | SD of selected hits in each charge bin; not the uncertainty on the mean. |
| Fit weights | SEM of each profile mean; not the SD. |
| Slope band | 16–84% range from 1000 bootstrap linear fits with fixed bin edges. |
| AWC bars / walk band | Propagate the slope replicas through a new AWC-plus-offset alignment to the same fixed reference curve. |

These definitions refer to the original displayed bands and bars. The next table additionally propagates the run-4399 quadratic covariance. Cross-run response transfer, clock/fine-time calibration, selection/model choices and cross-channel correlations remain outside both prescriptions. Neither is a total calibration uncertainty or a timing-resolution estimate.

# MCP-fit uncertainty and charge weighting

For each slope-bootstrap replica, propagate the correlated MCP coefficient covariance through the AWC-plus-offset fit. The Gaussian mixture gives the 16–84% interval; central AWC values and comparison curves are unchanged.

All table values are in $\mu$A. "Both runs" adds MCP-fit statistics, not cross-run or model systematics. The original plots retain slope-only bands.

| Channel | AWC | Slope only | Both runs | Max. weight shift |
|---|---|---|---|---|
| 133 | 36.03 | [32.73, 44.75] | [32.48, 45.72] | 4.11 |
| 154 | 24.25 | [23.92, 24.86] | [22.98, 25.84] | 0.85 |
| 137 | 18.33 | [18.12, 18.61] | [17.91, 18.80] | 2.05 |
| 150 | 20.89 | [20.65, 21.15] | [20.41, 21.38] | 1.13 |

Maximum weighting shift: largest $|\Delta\mathrm{AWC}|$ using uniform $1/Q$ or either run's charge population instead of uniform Q. Same overlap and free offset; these shifts are not statistical errors or added in quadrature.

All 21 overlap channels were checked; channel 238 has no overlap. A shared reference-time offset affects only the fitted offset, not AWC.

**Review N2-4:** Requested audit complete. Accept conditional statistical intervals with separate weighting sensitivity; a total uncertainty budget remains outside this check.

## Result and next validation step

- A dual-threshold measurement provides a charge-dependent slope proxy; an inverse-slope function can be aligned to independent MCP walk curves.
- Restricting alignment to jointly measured charge support avoids determining AWC from extrapolation.
- Reference-calibrated AWC varies between channels. A straight line against $1/c_1$ is insufficient for the explored subset.
- The present calibration still uses the MCP to determine AWC. It does not establish reference-free AWC extraction by itself.
- An internal predictor must be tested on held-out bars/configurations with transfer and target uncertainties included before use without MCP.

The following backup explores such a predictor; it is not a final performance claim or a demonstration with cosmic rays.

# Backup: an internal predictor for channel AWC

For $W = A/S$ and $S = c_0 + c_1 Q$,

$$-\frac{dW}{d\ln Q} = \frac{A\, c_1 Q}{S(Q)^2}.$$

If this local walk derivative is approximately shared between channels at a prescribed typical charge $Q_*$, a candidate model is

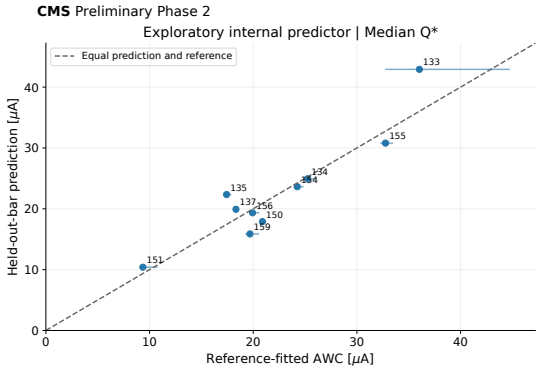
$$A_{\mathrm{pred}} = K\, \frac{S(Q_*)^2}{c_1 Q_*}.$$

Here $Q_*$ is the median charge of the run-4076 quality- and peak-right-selected sample. It is computed once per channel; $A_{\mathrm{pred}}$ remains constant when correcting individual events.

The common $K$ is learned from MCP-calibrated training channels. Predicting a new channel uses internal features, but the model development is reference-based.

**Review N2-7:** Shared local derivative is a hypothesis. Test operating-condition dependence and avoid selecting $Q_*$ only after inspecting prediction scores.

# Backup: predict a bar excluded from the model fit




Each point compares reference-fitted AWC with a prediction made after excluding that channel's entire bar from the fit of $K$. Ten upstream channels span six held-out bar groups.
The RMS prediction error is 3.22 $\mu$A; the RMS fractional error is 14.1%.
Horizontal intervals are conditional reference-AWC intervals, not predictive intervals.

**Review N2-7:** Small selected sample; feature/target correlations and full prediction uncertainties remain unpropagated.

Exploratory median-$Q_*$ candidate. The dashed line indicates equality, not a fitted result. No blanket 10% precision claim is made.

# References

[1] CMS Collaboration, *A MIP Timing Detector for the CMS Phase-2 Upgrade*, CERN-LHCC-2019-003, CMS-TDR-020 (2019). CERN document record.

[2] E. Albuquerque et al., *TOFHIR2: The readout ASIC of the CMS Barrel MIP Timing Detector* (2024). arXiv:2404.01208.

[3] W. S. Cleveland, *Robust Locally Weighted Regression and Smoothing Scatterplots*, J. Am. Stat. Assoc. **74** (1979) 829–836. doi:10.1080/01621459.1979.10481038.

[4] B. Efron, *Bootstrap Methods: Another Look at the Jackknife*, Ann. Statist. **7** (1979) 1–26. doi:10.1214/aos/1176344552.

These sources provide detector/readout context and methodological background. The run-specific selections, implementations and numerical results are those of this study, not measurements or prescriptions taken from these references.

# Review queue before publication

| ID | Decision or missing evidence |
|---|---|
| N2-3 | Model adequacy remains open; sensitivity audit saved separately. |
| N2-4 | Audit complete; accept statistical scope and weighting-sensitivity reporting. |
| N2-6 | Freeze channel eligibility independently of observed trend; publish exclusions. |
| N2-7 | Decide whether internal prediction remains backup or gains enough validation for a main result. |

Sources: supplied `Note2.pdf`; extended profile snapshots and the dated measured-overlap study. All plotted tables and checksums accompany this draft.

Primary-source references are included. Official template, DP metadata and cover handling still require convener confirmation. This review slide is removed from a public release.